

Basics of Electronics and Components

0. Introduction

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Details about the lecture, exercise and the exam

1	Scalar and vector fields
	Scalar fields, vector fields, vector calculus, Del-operator
2	Electric field
	Maxwell equations, static electric field, Gauss law, field applications
3	Magnetic field
	Static magnetic field, magnetization, magnetic circuit, time varying field
4	Semiconductor Physics and Diode Characteristics
	Electronic band structure, semiconductor junctions, Semiconductor diodes
5	Transistor Devices and Transistor Circuits
	Bipolar junction transistor and BJT circuits, Operational amplifier

Learning objectives

Students will be able to:

- define the principles of vector calculus (scalars, vectors, fields, del operator, etc.) and apply them in calculations
- state the physical principles governing electric and magnetic fields
- construct, calculate, and sketch electric and magnetic field distributions
- explain the physical principles of semiconductor devices
- list and compare the properties various transistor types
- design simple diode and transistor circuits and analyse their behaviour

Basics of Electronics and Components

Course details

Language

Lecture

- Spoken language: **German**
- Slides:
 - └ **English**
 - └ German: ASAP after automated, structure preserving translation is available

Exercises

- **German**
- English exercise sets available on the IRO2/BEC e-learning #32813

Exam

- **German**

Course details

Lecture

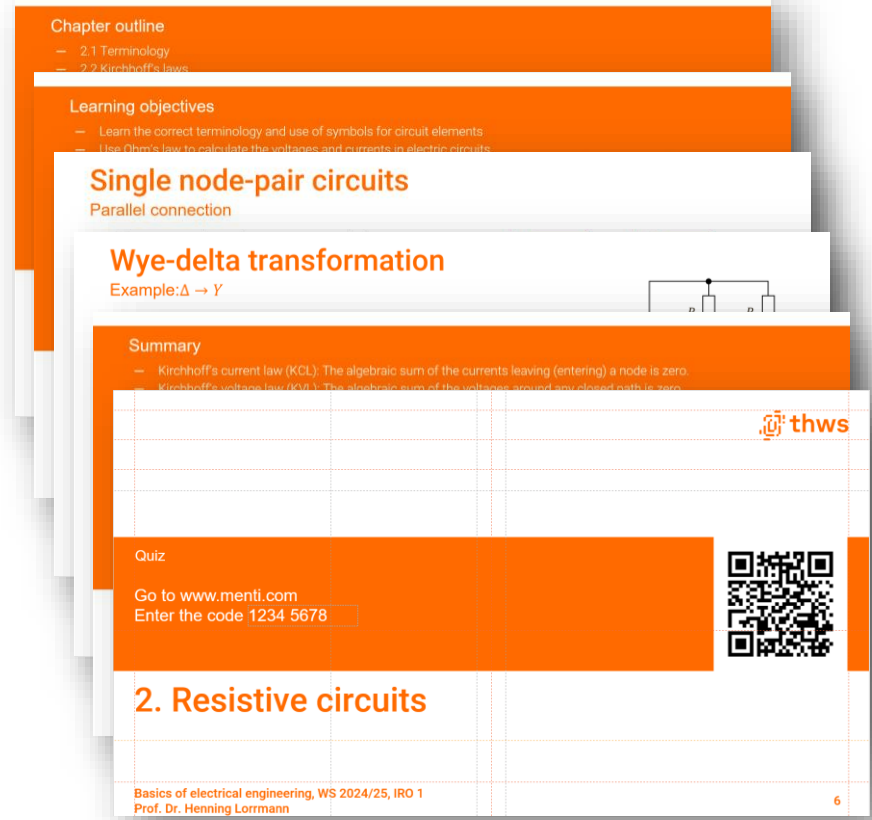
— Slide Structure

- 7 Script-like format with one concept per slide
- 7 Student version: exercises with fill-in boxes
- 7 Instructor version: complete solutions (available after quick test on the e-learning)

— Chapter structure

- 7 Chapter outline
- 7 Learning objectives
- 7 Learning content
- 7 Step-by-step worked examples
- 7 Concept summary
- 7 Interactive elements (quizzes, peer instruction)

Active participation throughout the course is essential. Students who choose not to engage at least **10 h/week** should be aware that their chances of passing the exam are extremely low.



Course details

Moodle tests

- Tests use **uncertainty-based marking (CBM)**
- After each answer, choose how certain you are on a 3-point scale: $C = 1$ (low), $C = 2$ (mid), $C = 3$ (high).
 - ⌋ If you are **correct**, $C = 1, 2, 3$ give you 1, 2, 3 marks.
 - ⌋ If you are **wrong**, you lose 0, 2, or 6 marks at $C = 1, 2, 3$.
- Tests give a total grade between 0 and 30 (up to 300 %).
- Complete slides are unlocked with a grade of at least 10 (100 %).
- *Example: correct at $C = 2 \rightarrow +2$ marks; wrong at $C = 3 \rightarrow -6$ marks.*

Certainty level	$C = 1$	$C = 2$	$C = 3$
Mark if correct	1	2	3
Mark if wrong	0	-2	-6
Probability correct	< 67%	67-80%	> 80%
Description	Unsure	Mid	Quite sure



Highly confident wrong answers are heavily penalized → judge your certainty carefully before choosing C .

Highly confident wrong answers usually **signal a misconception**, so review the feedback.

Course details

Exercise

- New exercise sets will be uploaded **weekly**.
- **Bonus program** (APO § 29):
 - 7 Presenting one exercise at the board earns a **one-step grade improvement** (e. g. 2.0 → 1.7) for a correct and well-explained solution.
 - 7 **Registration** per E-mail or before/after class required one week in advance after the exercise session.

Completing at least 75% of all assigned exercises is a prerequisite for success. Without consistent practice, the probability of passing the exam approaches zero — in such cases, registering for the exam is strongly discouraged.



Course details

Exam

— Exam format

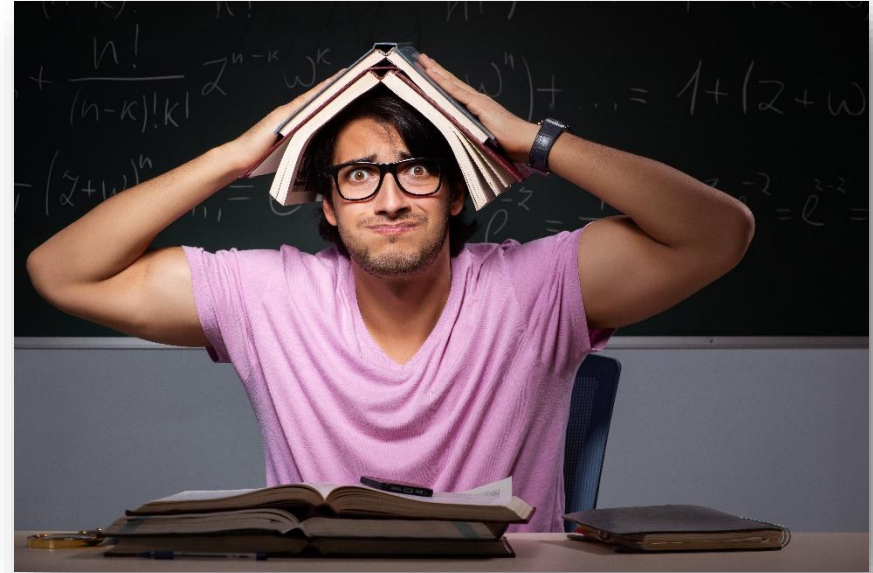
- 7 Written exam
- 7 **Language:** German
- 7 **Duration:** 90 min
- 7 **Authorized aids:** formulary provided by examiner, one A4 sheet with handwritten notes
- 7 Regulations and aids are **subject to change**

— Important advice

- 7 Start preparing during the lecture period; there is little time afterward
- 7 Actively join the **exercise** sessions to gain experience and exam practice

— Questions about the exam are answered only

- 7 in the lecture,
- 7 exercise sessions,
- 7 or the online forum



Course details

Literature

Calculus

- Paul Dawkins – Calculus III

<https://tutorial.math.lamar.edu/Classes/CalcIII/CalcIII.aspx>

Electromagnetism

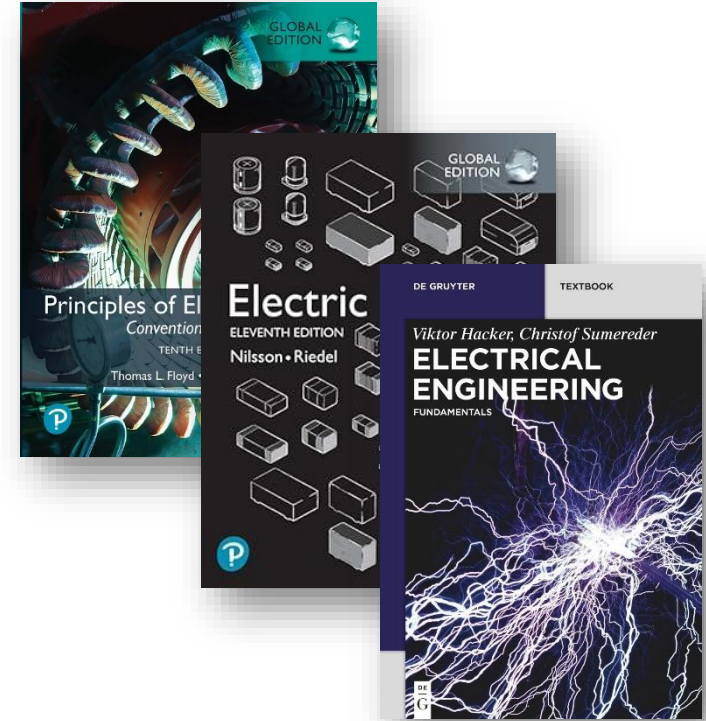
- The Feynman Lectures on Physics, Volume II

<https://www.feynmanlectures.caltech.edu/>

- D. J. Griffiths – Introduction to Electrodynamics
- S. Paul – Grundlagen der Elektrotechnik 2:
Elektromagnetische Felder und ihre Anwendung

Electronics and components

- T. L. Floyd – Principles of Electric Circuits
- J. Nilsson and S. Riedel – Electric Circuits
- Viktor Hacker, Christof Sumederer – Electrical Engineering



Organizational information

Organizational matters

Lecture and exercise details

Lecture (4 SWS)

- Wednesday, 10:00 – 11:30
 - ⌋ Location: Campus Konrad Geiger
 - ⌋ Room: 9.E.24
- Thursday, 14:15 – 15:45
 - ⌋ Location: Campus Konrad Geiger
 - ⌋ Room: 9.E.24

Exercise (2 SWS)

- Wednesday, 11:45 – 13:15
 - ⌋ Location: Campus Konrad Geiger
 - ⌋ Room: 9.E.24
- **Text exam:**
 - ⌋ Tentative date 13.05.2026
 - ⌋ Chapters 1 – 3
- **BRO/IRO joint lecture:**
 - ⌋ 30.06.2026
 - ⌋ 07.07.2026

Tuesday	Wednesday	Thursday
	18.03.2026	19.03.2026
	25.03.2026	19.03.2026
	01.04.2026	02.04.2026
	08.04.2026	09.04.2026
	15.04.2026	16.04.2026
	22.04.2026	23.04.2026
	29.04.2026	30.04.2026
	06.05.2026	07.05.2026
	13.05.2026	14.05.2026
	20.05.2026	21.05.2026
	27.05.2026	28.05.2026
	03.06.2026	04.06.2026
	10.06.2026	11.06.2026
	17.06.2026	18.06.2026
	24.06.2026	25.06.2026
30.06.2026	01.07.2026	02.07.2026
07.07.2026	08.07.2026	09.07.2026

Lecture
BRO/IRO joint lecture
Exercise
Free
Test exam

Organizational matters

E-Learning

- Lecture notes and exercises will be published via e-learning:
- [BRO2: Grundlagen der Elektronik / Bauelemente](#)
 - └ Course ID: 32812
 - └ Enrolment key: GDEBSS26
- Please check regularly for updated versions!
- Version structure: X.Y.Z
 - └ X is the major version
 - └ Y is the minor version
 - └ Z is the patch version

